

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

VIDEO GAME SALES DATA ANALYSIS AND VISUALIZATION USING PYTHON AND POWER BI

Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework of Data Visualization and Analytics (FDVA), analyses a Video Game Sales dataset using Python and Microsoft Power BI. The dataset is stored in a file named vgsales.csv and contains 16,598 records with 11 columns: rank, game name, platform, year of release, genre, publisher, and sales in millions of copies for North America, Europe, Japan, other regions and the world. The raw data had missing values in the Year (271) and Publisher (58) columns, the Year was stored as a decimal number, and two games were listed twice. Python with Pandas was used to correct these problems, create new columns such as regional sales share, Dominant_Region, Decade and Sales_Category, and save a cleaned file with 16,596 rows and 19 columns. Matplotlib and Seaborn were then used to create eight charts, including the yearly sales trend, genre-wise and platform-wise sales, a genre-by-region heatmap and a correlation heatmap. An interactive dashboard was also developed in Power BI, with charts by year, genre, platform and region, a treemap, a KPI visual for target sales and a regional sales pie chart. The analysis showed that total global sales were about 8,920 million copies, that sales peaked in 2008, that Action was the best-selling genre, that the PS2 was the best-selling platform, and that Japan prefers Role-Playing games unlike every other region. The project shows how preprocessing, charts and dashboards turn raw sales records into clear information that can support business decisions.

1. Data Preprocessing

The dataset (vgsales.csv) has 16,598 rows and 11 columns. It was cleaned in Python using Pandas, as shown below.

1.1 Loading and inspecting the data

```
df = pd.read_csv("vgsales.csv")
df.shape        # (16598, 11)
df.info()
```

Rank is an integer, the five sales columns are decimals in millions of copies, and Name, Platform, Genre and Publisher are text. Year was read as a decimal number (for example 2006.0) because it has missing values.

1.2 Missing values and duplicates

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
print("Same game listed twice:",
      df.duplicated(['Name', 'Platform', 'Year']).sum())
```

Year (271) and Publisher (58) had missing values. No fully duplicate rows were found (0), but 2 games (Madden NFL 13 on PS3 and Wii de Asobu: Metroid Prime on Wii) appeared twice with the same name, platform and year.

1.3 Fixing data types and filling missing values

```
df['Year'] = pd.to_numeric(df['Year'], errors='coerce').astype('Int64')
df['Publisher'] = df['Publisher'].fillna('Unknown')
df = df.drop_duplicates(['Name', 'Platform', 'Year'])
```

Year was converted to a whole-number type that allows missing values. Missing publishers were filled with “Unknown”. Rows with no Year were kept for overall totals but left out of the year-wise charts, because guessing a release year would give wrong trends. The 2 repeated games were removed, leaving 16,596 rows.

1.4 Standardising text values and checking sales

```
for col in ['Name', 'Platform', 'Genre', 'Publisher']:
    df[col] = df[col].astype(str).str.strip()

regional = ['NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales']
diff = (df[regional].sum(axis=1) - df['Global_Sales']).abs()
print(diff.max())                       # 0.02
print((df['Global_Sales'] <= 0).sum())  # 0
```

Extra spaces were removed from the text columns. The four regional sales add up to Global_Sales for every game (the largest difference is 0.02 million, which is only rounding), and no sales value is zero or negative. The years 2017 and 2020 have only 4 records between them, so they are incomplete.

1.5 New columns and export

```
for r in regional:
    df[r.replace('_Sales', '_Share')] = df[r] / df['Global_Sales']
df['Dominant_Region'] = df[regional].idxmax(axis=1).str.replace('_Sales', '')
df['Decade'] = (df['Year'] // 10 * 10).astype('string') + 's'
df['Sales_Category'] = pd.cut(df['Global_Sales'], [-1, 0.1, 1, 5, 1000],
    labels=['Niche', 'Moderate', 'Hit', 'Blockbuster'])
df.to_csv("Cleaned_vgsales.csv", index=False)   # shape (16596, 19)
```

Eight columns were added (Regional_Total, NA_Share, EU_Share, JP_Share, Other_Share, Dominant_Region, Decade and Sales_Category). The cleaned dataset with 16,596 rows and 19 columns was saved and used for the visualizations and the dashboard.

2. Data Visualization Using Python

Eight charts were created from the cleaned data using Pandas, Matplotlib and Seaborn (sns.set_theme(style="whitegrid")). Each chart is explained briefly below.

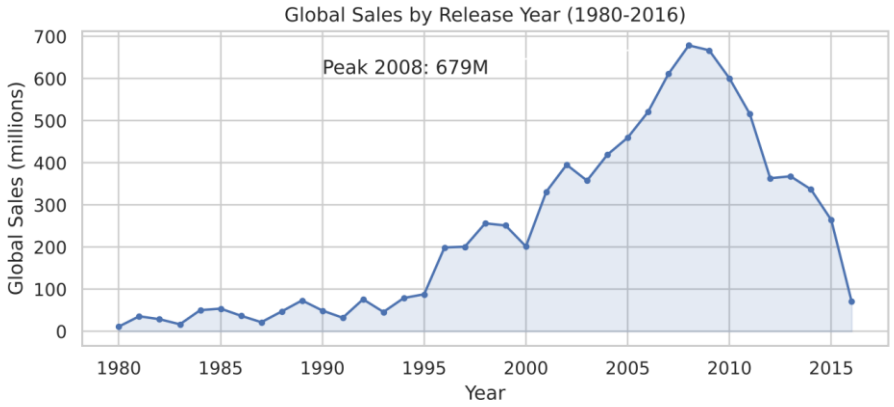


Figure 2.1: Global sales by release year (line chart)

Result: Sales rose steadily from the mid-1990s and peaked in 2008 (679M copies), then fell after 2011. The sharp drop after 2015 is partly because recent years are incomplete in the dataset.

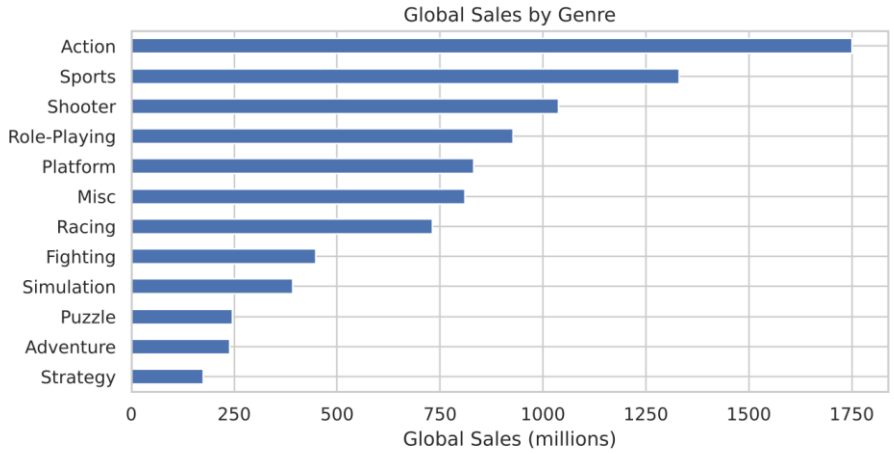


Figure 2.2: Global sales by genre (bar chart)

Result: Action is highest (1,751M), followed by Sports (1,331M) and Shooter (1,037M); Strategy is lowest (175M).

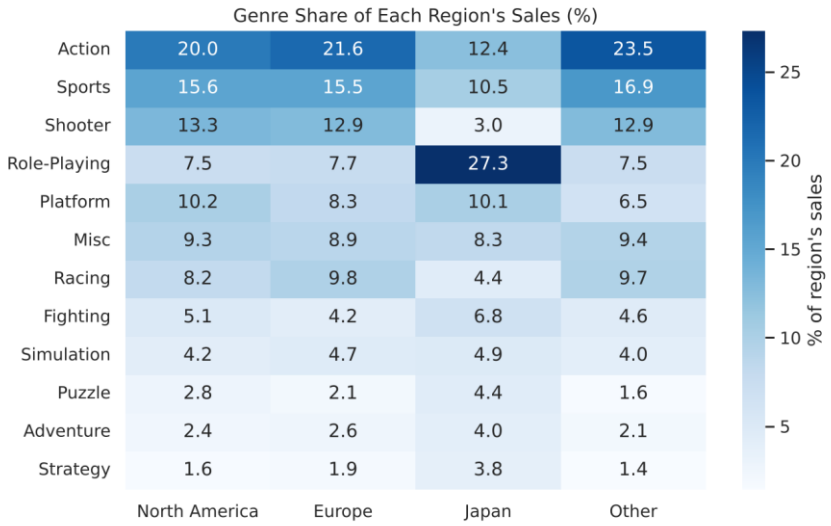


Figure 2.3: Genre share of each region's sales (sns.heatmap)

Result: Role-Playing is 27.3% of Japan's sales but under 8% in North America and Europe; Shooter is only 3.0% in Japan.

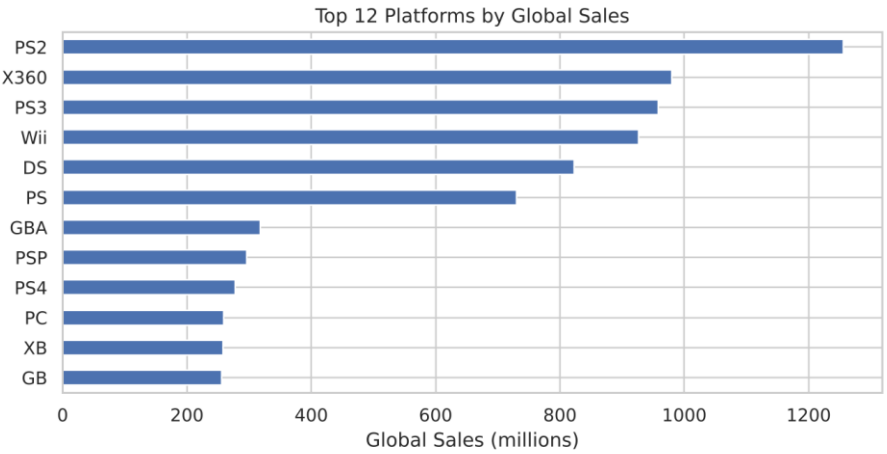


Figure 2.4: Top 12 platforms by global sales

Result: PS2 leads (1,256M), followed by X360 (980M), PS3 (958M) and Wii (927M).

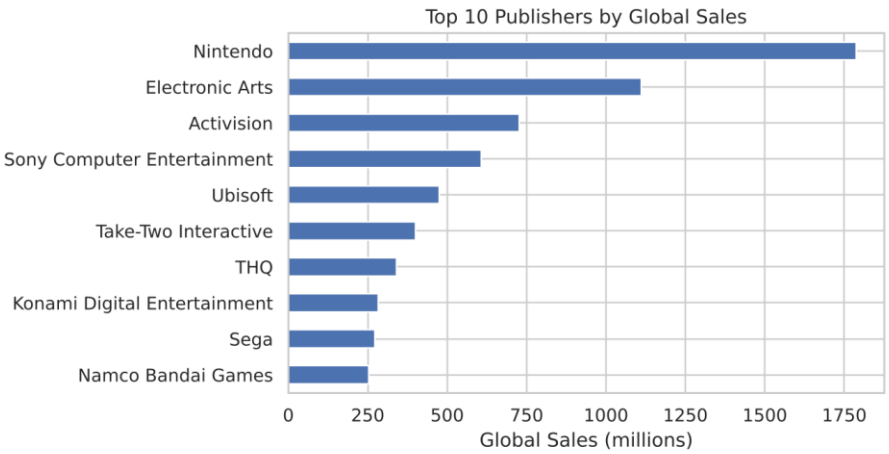


Figure 2.5: Top 10 publishers by global sales

Result: Nintendo is far ahead (1,787M), followed by Electronic Arts (1,110M) and Activision (727M).

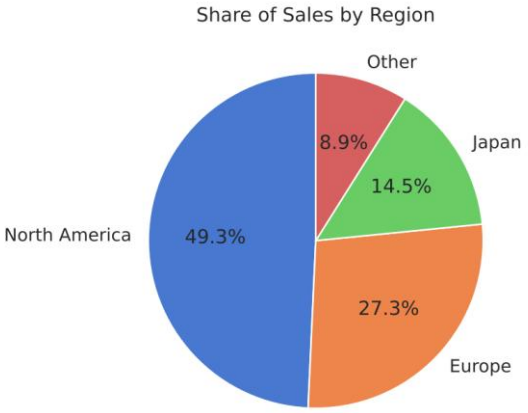


Figure 2.6: Share of sales by region (pie chart)

Result: North America 49.3%, Europe 27.3%, Japan 14.5%, Other 8.9%.

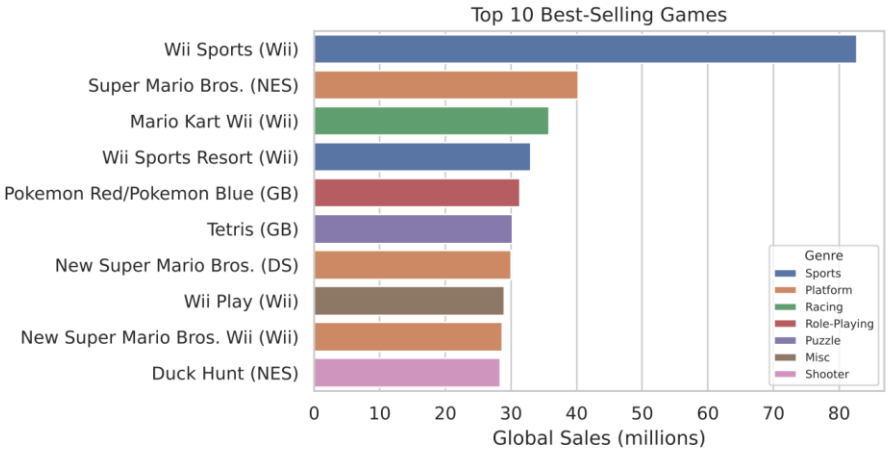


Figure 2.7: Top 10 best-selling games (sns.barplot)

Result: Wii Sports is far ahead (82.7M); Super Mario Bros. (40.2M) and Mario Kart Wii (35.8M) follow, and all ten are Nintendo titles.

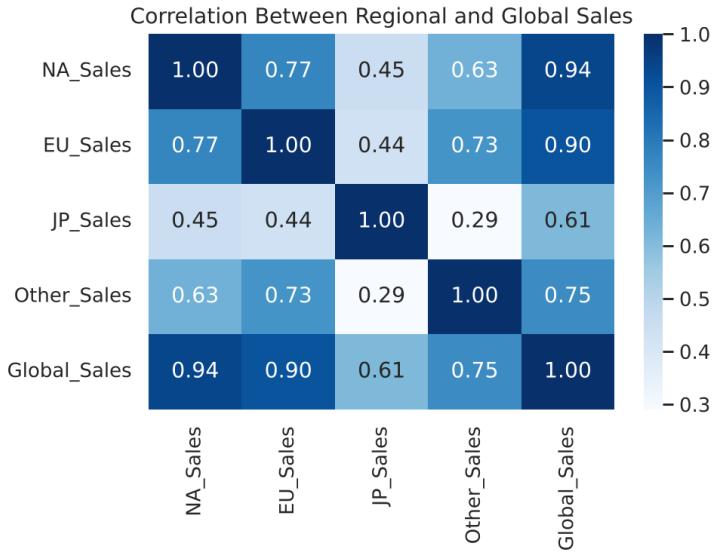


Figure 2.8: Correlation heatmap

Result: NA_Sales is most strongly related to Global_Sales (0.94), followed by EU_Sales (0.90); JP_Sales is the weakest (0.61).

Overall, a few genres, platforms and publishers account for most sales, North America is the largest market, and Japan behaves differently from the other regions.

3. Power BI Dashboard

Power BI was used to build a one-page interactive dashboard from the video game sales data.

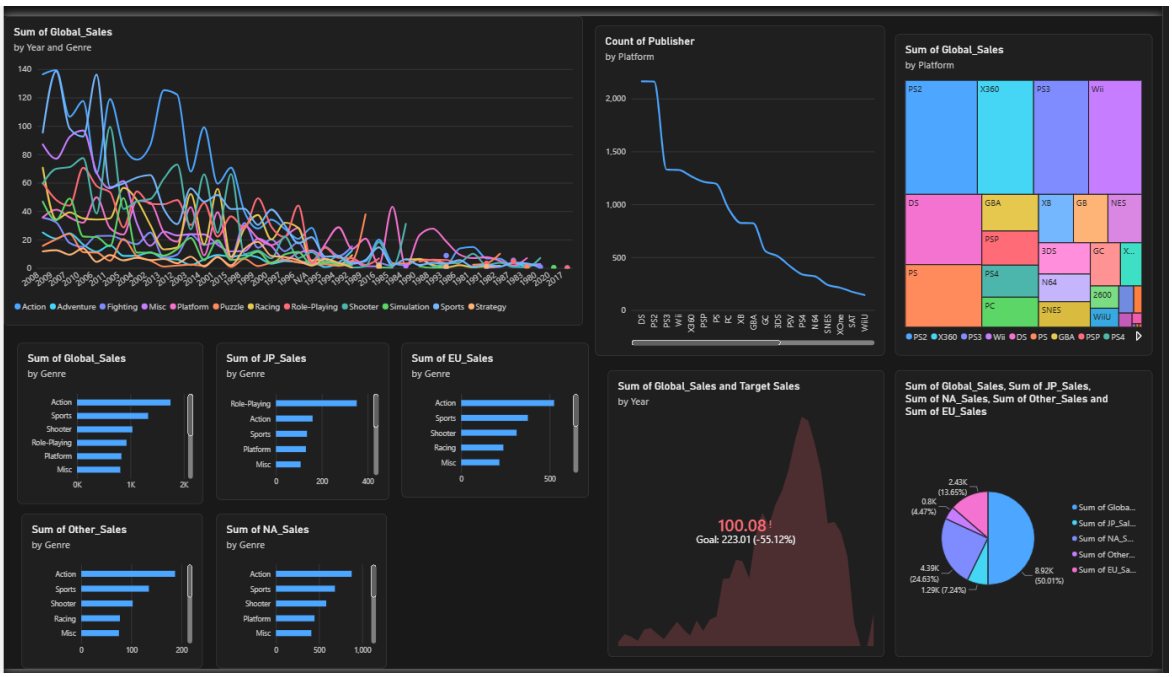


Figure 3.1: Video Game Sales Dashboard in Power BI

- **Line charts:** Sum of Global_Sales by Year and Genre, with one line per genre, and Count of Publisher by Platform, which shows how many titles were released on each platform.
- **Treemap:** Sum of Global_Sales by Platform, where the size of each box shows the platform's share of sales.
- **Bar charts:** Sum of Global_Sales, NA_Sales, EU_Sales, JP_Sales and Other_Sales by Genre, one chart for each region.
- **KPI and pie chart:** a KPI visual compares global sales with Target Sales (100.08 against a goal of 223.01, which is 55.12% below target), and a pie chart shows Global 8.92K, NA 4.39K, EU 2.43K, JP 1.29K and Other 0.8K.
- **Insights:** Action is the top genre in every region except Japan, where Role-Playing leads; PS2, X360, PS3, Wii and DS have the largest share of sales; DS and PS2 have the most titles (about 2,160 each).
- **Note:** the year axis of the line chart is sorted by sales value and not in time order, and "Count of Publisher" counts game records, not different publishers.